Professors Stonier and Hague, "Even, with a given technology, a firm can 'learn' to produce at a lower unit cost the longer the period of time that has elapsed since a previous observation and the greater the aggregate amount of that product that has consequently been made. It may be, therefore, that technological change is not the only reason why long-run cost curves are L-shaped rather than U-shaped.[16]

We have stated above that according to empirical evidence long-run average cost after the initial rapid fall, either remains constant or declines throughout; it does not rise. That is, long-run average cost does not turn up as is required if it is *U*-shaped. Professor G.A. Smith[17] who has examined empirical evidence in this connection has concluded that with a very large size of firm, labour costs, assembly costs and distribution costs increase very much and therefore large-sized plants *with increasing average cost are not set up in actual practice* and therefore empirical evidence cannot assess the cost situation in them. Therefore, according to Professor Smith, empirical evidence does not refute the U-shaped nature of long-run average cost curve. Walters[18] after examining the empirical studies by Dean and Johnston reached the conclusion that, there is no large body of data which convincingly contradicts the hypothesis of a *U*-shaped long-run average cost curve and the fruitful results which depend on it".

NUMERICAL EXERCISES

Exercise 1. *Suppose a firm faces a cost function of* $C = 8+4q+q^2$

(i) What is the firm's fixed cost?

(ii) Derive an expression for the firm's variable cost and marginal cost.

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Solution :

(*i*) As fixed cost of the firm does not vary with output, the term in the given cost function which has no output (*q*) variable will be the fixed cost. From the given cost function it is evident that fixed cost is 8.

(*ii*) Total variable cost (*TVC*) = *TC* – *TFC*

$$= (8+4q+q^2)-8 = 4q+q^2$$

and,
$$AVC = \frac{TVC}{Q} = \frac{4q+q^2}{q} = 4+q$$

Marginal cost is the first derivative of total cost function or total variable cost function. Thus, taking the first derivative of the total variable cost function we have

$$MC = \frac{d(TVC)}{dq} = 4+2q$$

Exercise 2. *Given the total cost function* $TC = 1000 + 4Q + 0.5Q^2$

Determine the functions for TFC, TVC, MC, AVC and AC

Solution :

$$TC = 1000 + 4Q + 0.5Q^2$$

Total fixed cost is independent of level of output to be produced.

Therefore, $TFC = 1000$

$$TVC = TC - TFC$$

Therefore, $TVC = 4Q+0.5Q^2$

16. Stonier and Hague, *op. cit.*, p. 141.
17. C.A. Smith, Empirical Evidence of Economies of Scale, printed in *Theory of Firm* (Penguin)
18. A. Walters, Production and Cost Fucntions, An Econometric Survey, *Econometrica* Vol. 31, 1963, p. 52.

$$AVC = \frac{TVC}{Q}$$

$$\therefore \quad AVC = \frac{4Q + 0.5Q^2}{Q} = 4 + 0.5Q$$

$$MC = \frac{\Delta TVC}{\Delta Q} = 4 + 1.0Q$$

or $$MC = 4 + Q$$

$$AC = AFC + AVC$$

$$AFC = \frac{TFC}{Q} = \frac{1000}{Q}$$

Therefore $$AC = \frac{1000}{Q} + 4 + 0.5Q$$

Exercise 3. *We give below short-run total cost function* :

$$TC = 100 + 50Q - 12Q^2 + Q^3$$

where TC is total cost and Q is level of output.

(*i*) *Determine* : (*a*) *total fixed cost function* (b) *total variable cost function*, (c) *average variable cost function*, (*d*) *marginal cost function.*

(*ii*) *Calculate total cost, ATC, AVC and MC when the firm produces 10 units of output.*

(*iii*) *Calculate the level of output at which AVC is minimum.*

Solution. (*a*) From the total cost function it is evident that when level of output is zero, the firm's cost will be Rs. 100. Thus, $TFC = 100$

(*b*) If we deduct Rs. 100 from the total cost function we obtain the total variable cost function.

Thus, $$TVC = TC - TFC$$

$$\therefore \quad TVC = 50Q - 12Q^2 + Q^3$$

(*c*) If we divide the *TVC* function by output (Q) we get *AVC* function. Thus,

$$AVC = \frac{50Q - 12Q^2 + Q^3}{Q}$$
$$= 50 - 12Q + Q^2$$

(*d*) Marginal cost is the first derivative of total variable cost function. Thus,

$$MC = \frac{dTVC}{dQ} = 50 - 24Q + 3Q^2$$

(*ii*) Given total output (Q) equal to 10, we have

$$TC = 100 + 50(10) - 12(10)^2 + (10)^3$$
$$= 100 + 500 - 1200 + 1000$$
$$= 400$$

$$ATC = \frac{100}{Q} + 50 - 12Q + Q^2 = \frac{100}{10} + 50 - 12 \times 10 + (10)^2$$
$$= 10 + 50 - 120 + 100$$
$$= 40$$

$$AVC = 50 - 12Q + Q^2$$
$$= 50 - 12 \times 10 + 100$$
$$= 30$$

$$MC = 50 - 24Q + 3Q^2$$
$$= 50 - 240 + 300$$
$$= 110$$

(*iii*) The level of *AVC* is minimum at the level of output at which first derivative of *AVC* function is zero. Thus,

$$AVC = 50 - 12Q + Q^2$$
$$\frac{d(AVC)}{dQ} = -12 + 2Q = 0$$
$$2Q = 12$$
$$Q = 6$$

Thus at 6 units of output, *AVC* is minimum.

Exercise 4. *Given the following total cost function TC = 1500 + 15Q – 6Q² + Q³*

(i) Determine the total fixed cost for producing 1000 units of output and 500 units of output.

(ii) What is AFC at (a) 1000 units of output, and (b) 500 units of output

(iii) Determine TVC, AVC, MC and AC at 50 units of output.

Solution :

(*i*) *TFC* = 1500 irrespective the level of output

(*ii*) *AFC* at 1000 units of output :

$$AFC = \frac{TFC}{Q} = \frac{1500}{1000} = 1.5$$

AFC at 500 units of output

$$AFC = \frac{TFC}{Q} = \frac{1500}{500} = 3$$

(*iii*) $$TVC = TC - TFC$$

$$\therefore \quad TVC = 15Q - 6Q^2 + Q^3$$

If output is 50 units

$$TVC = 15 \times 50 - 6 \times (50)^2 + (50)^3$$
$$= 750 - 15000 + 125000$$
$$= 1,10,750$$

$$AVC = \frac{TVC}{Q} = \frac{1,10,750}{50} = 2215$$

$$MC = \frac{dTVC}{dQ} = 15 - 12Q + 3Q^2$$

MC with $Q = 50, = 15 - 600 + 7500 = 6915$

$$AC = AFC + AVC$$

AC with $Q = 50, = \frac{1500}{50} + 2215 = 30 + 2215 = 2245$